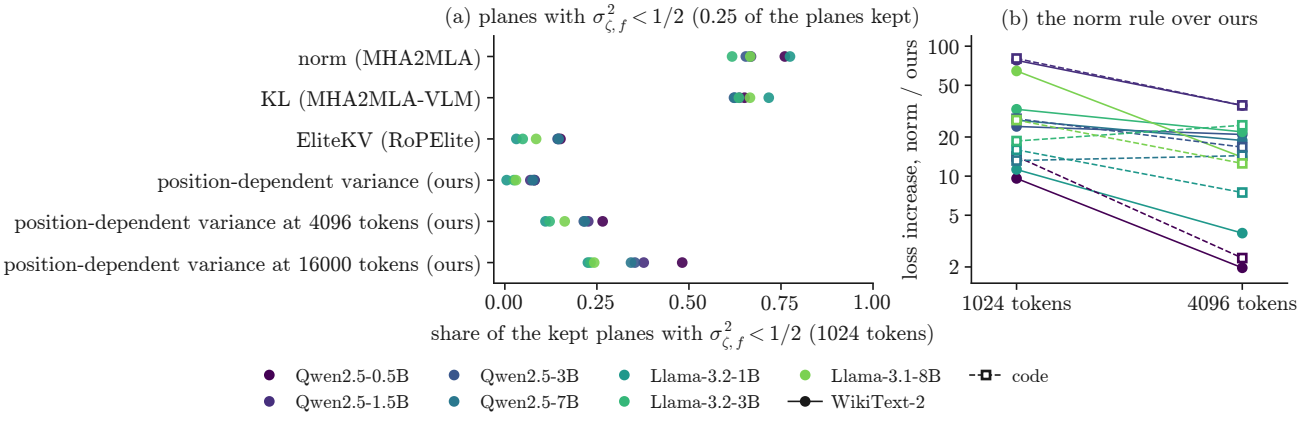
(a) planes with $\sigma^2_{\zeta,f} < 1/2$ (0.25 of the planes kept)
norm (MHA2MLA)
KL (MHA2MLA-VLM)
EliteKV (RoPElite)
position-dependent variance (ours)
position-dependent variance at 4096 tokens (ours)
position-dependent variance at 16000 tokens (ours)
0.00
0.25
0.50
0.75
1.00
share of the kept planes with $\sigma^2_{\zeta,f} < 1/2$ (1024 tokens)
(b) the norm rule over ours
loss increase, norm / ours
100
50
20
10
5
2
1024 tokens
4096 tokens
Qwen2.5-0.5B
Qwen2.5-1.5B
Qwen2.5-3B
Qwen2.5-7B
Llama-3.2-1B
Llama-3.2-3B
Llama-3.1-8B
WikiText-2
code